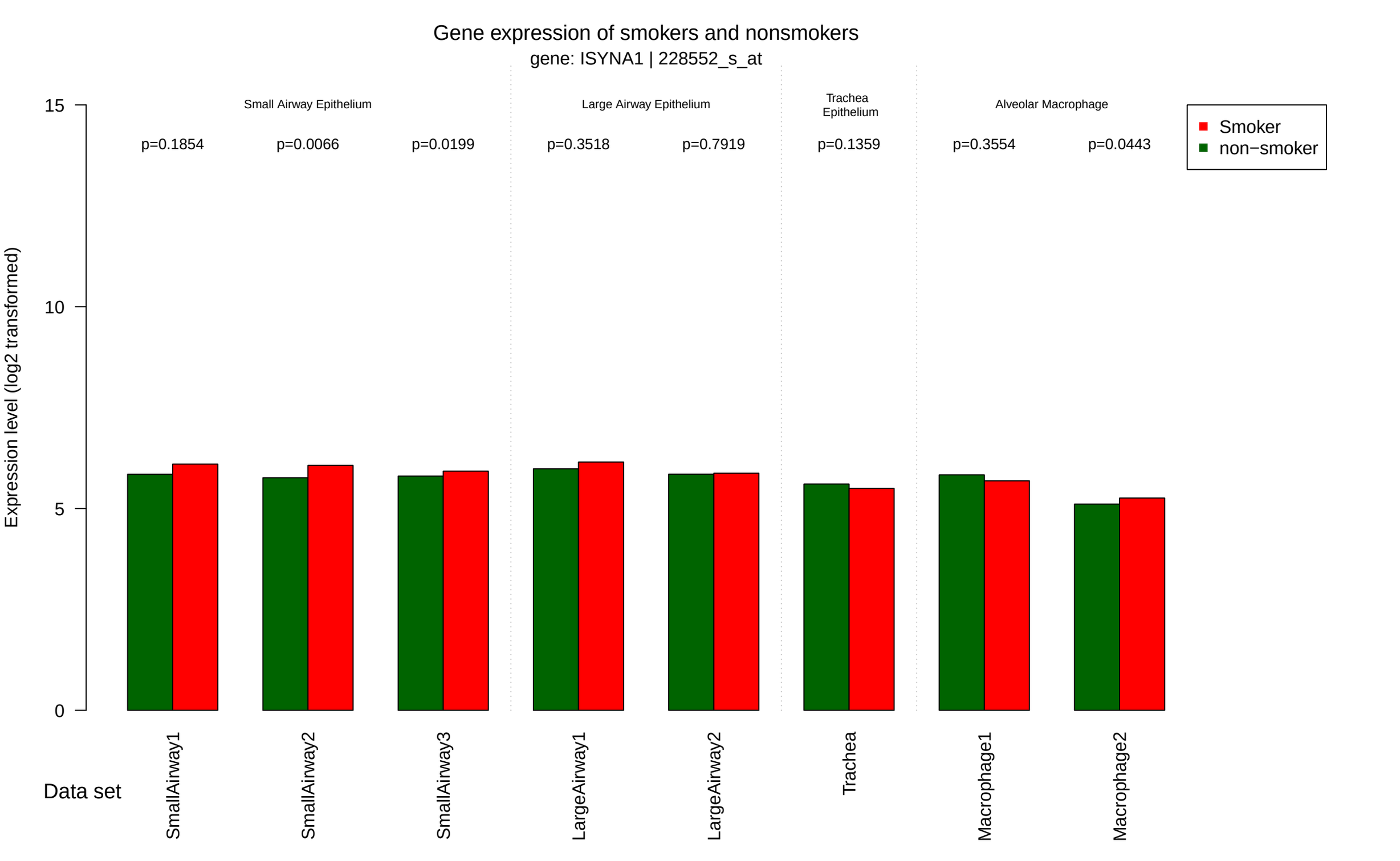
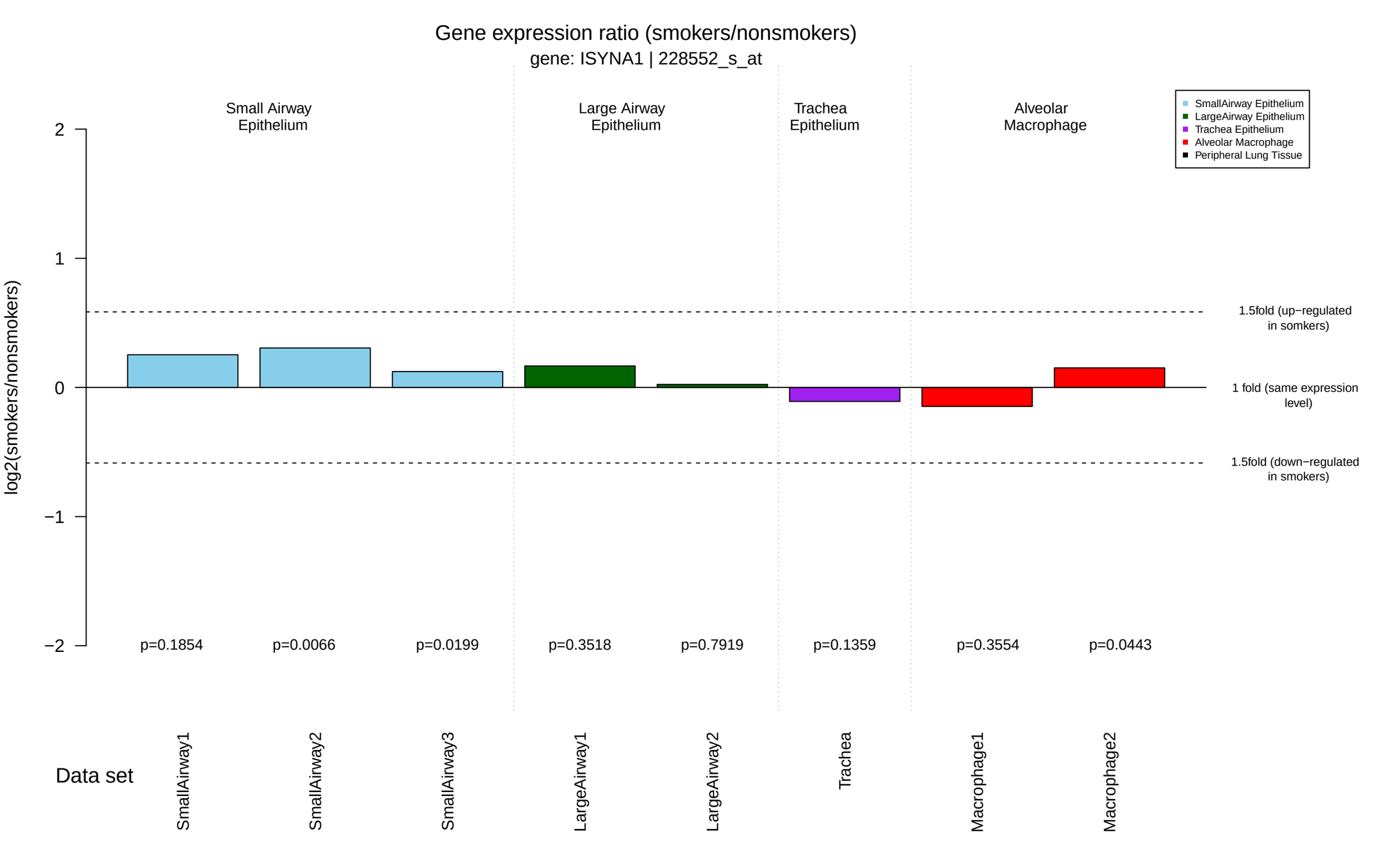
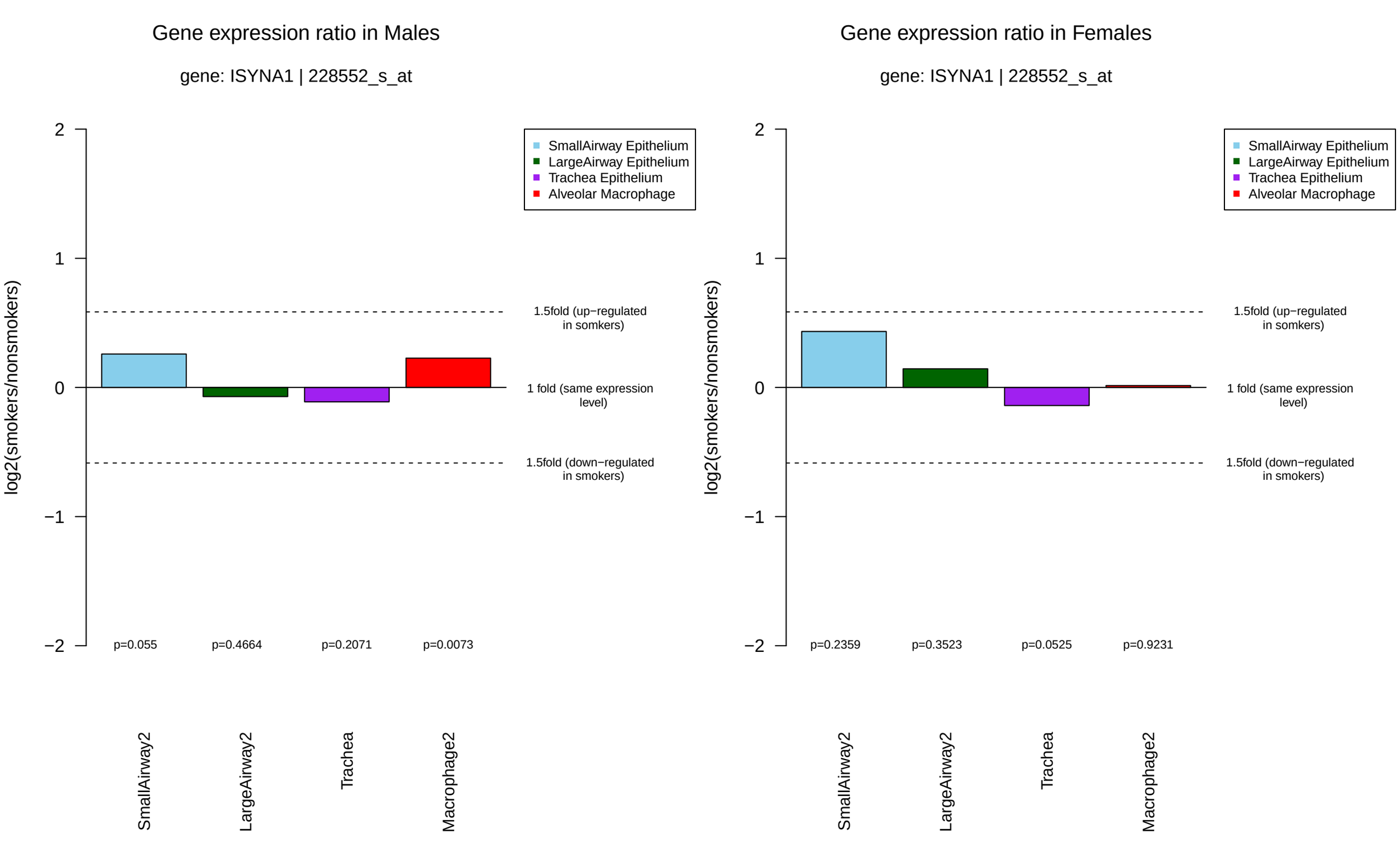
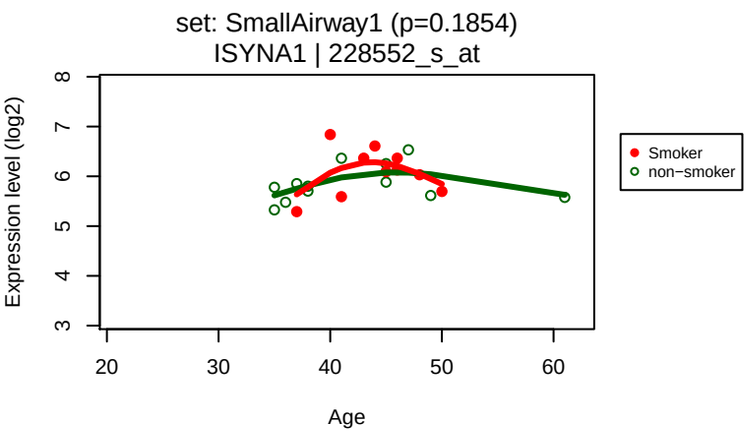
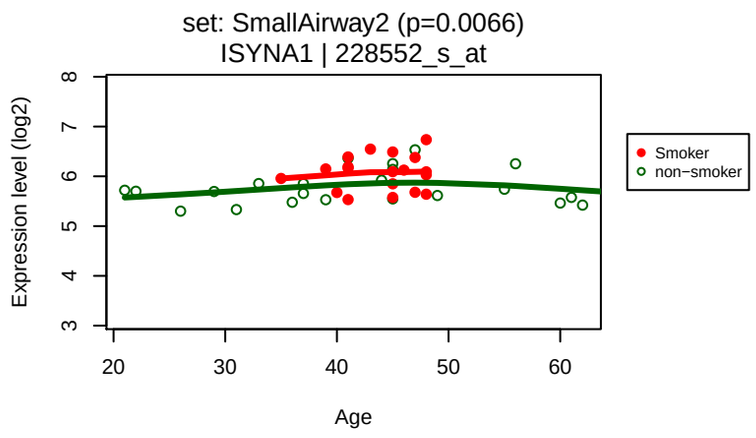
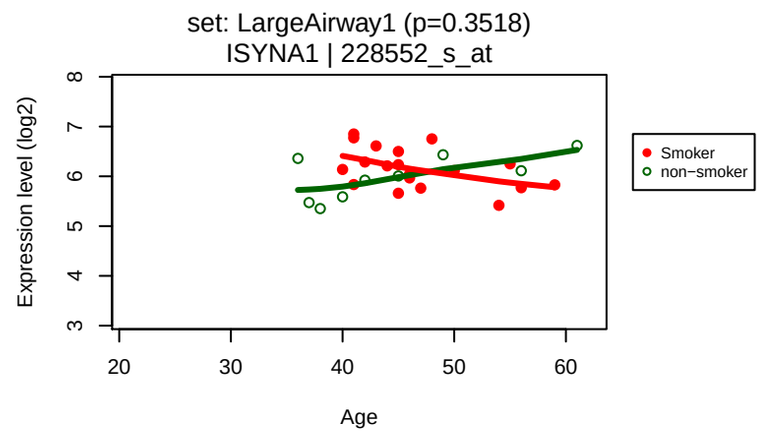

Gene expression of smokers and nonsmokers
gene: ISYNA1 | 228552_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1854
p=0.0066
p=0.0199
p=0.3518
p=0.7919
p=0.1359
p=0.3554
p=0.0443
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

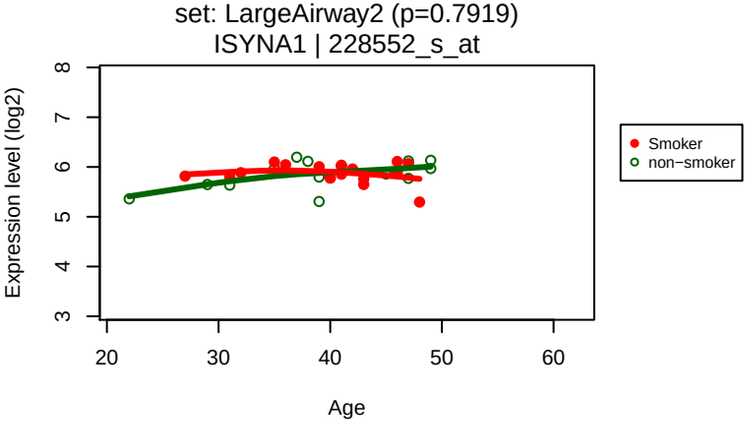

Gene expression ratio (smokers/nonsmokers)
gene: ISYNA1 | 228552_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1854
p=0.0066
p=0.0199
p=0.3518
p=0.7919
p=0.1359
p=0.3554
p=0.0443
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

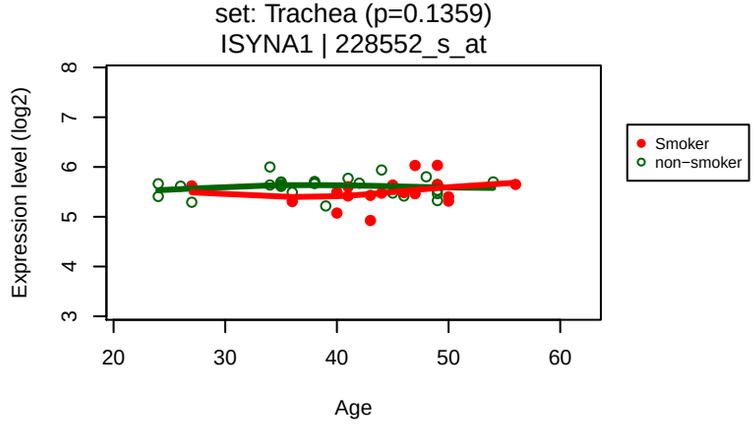

Gene expression ratio in Males
gene: ISYNA1 | 228552_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.055
p=0.4664
p=0.2071
p=0.0073
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ISYNA1 | 228552_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2359
p=0.3523
p=0.0525
p=0.9231
SmallAirway2
LargeAirway2
Trachea
Macrophage2

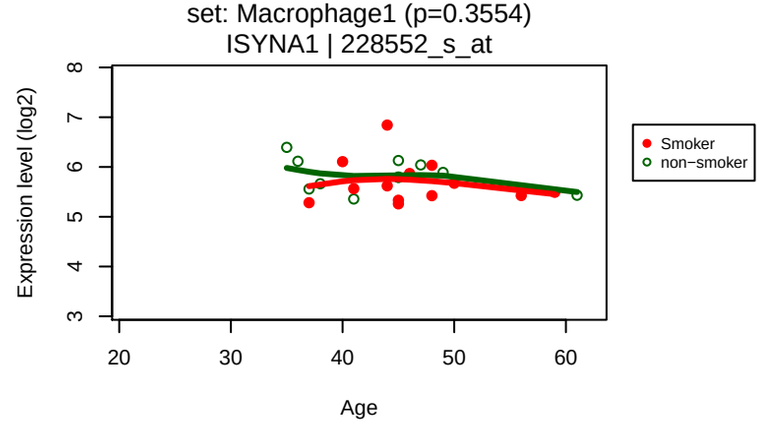

set: SmallAirway1 (p=0.1854)
ISYNA1 | 228552_s_at
Expression level (log2)
Age
Smoker
non−smoker

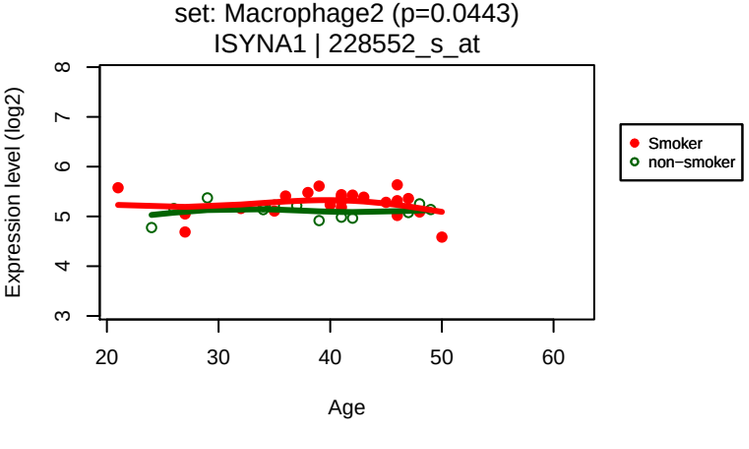

set: SmallAirway2 (p=0.0066)
ISYNA1 | 228552_s_at
Expression level (log2)
Age
Smoker
non−smoker

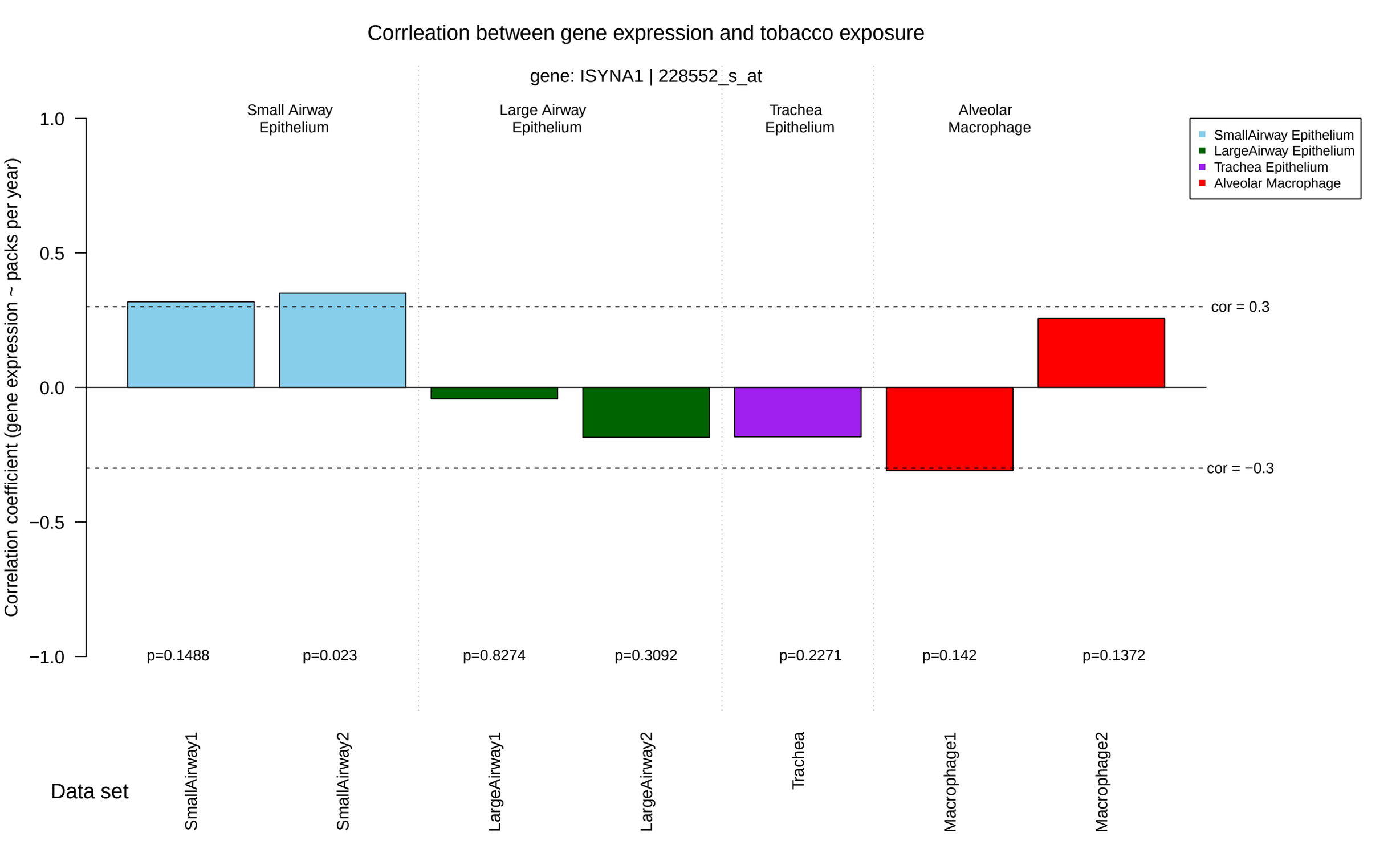

set: LargeAirway1 (p=0.3518)
ISYNA1 | 228552_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7919)
ISYNA1 | 228552_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1359)
ISYNA1 | 228552_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3554)
ISYNA1 | 228552_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0443)
ISYNA1 | 228552_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ISYNA1 | 228552_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1488
p=0.023
p=0.8274
p=0.3092
p=0.2271
p=0.142
p=0.1372
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2